

Guide: Where the Current Merge Table Cannot Help

Collatz proof project — September 5, 2026

The question

The previous result let us combine several verified merge rules after moving forward along an orbit. Could a fixed number of forward steps, followed by as many rules as needed, cover every remaining candidate? For the current table of 360 rules, the answer is no. There are systematic arithmetic gaps, even when rule-chain length is unlimited.

The construction

Choose any proposed forward horizon K . Take a seed N satisfying

$$N + 1 \text{ is divisible by } 2^{K+18}, \quad N \text{ is divisible by } 3^{11}.$$

These two requirements are compatible and give arbitrarily large seeds. The first requirement makes the first K steps odd. The second controls which reverse rules can apply to the resulting values.

The exact residue check shows that at each of these forward values, the only available rule undoes one odd step. Applying it repeatedly returns to N , where no listed rule applies. Stopping earlier leaves a value larger than N . No chain can get below the original seed.

What Lean verifies

Lean checks twelve complete arithmetic classes. On eleven of them, only the inverse odd rule can apply; on the initial class no rule can apply. These claims hold for arbitrarily large members of the classes, not just their smallest representatives. The finite checks use ordinary kernel reduction and are included in the theorem audit.

The technical paper then proves, in ordinary mathematics, that the construction above stays within those classes for any horizon K . That arbitrary-horizon deduction has not yet been formalized in Lean. The Python program independently reconstructs the classes and replays sample horizons, including 128, with several progression quotients.

What this rules out

Simply increasing a fixed forward-search limit and allowing longer chains from the same table cannot establish universal coverage of finite-prefix candidates. For every chosen limit, the construction supplies candidates that defeat it. The seed depends on the limit; this does not construct one orbit defeating every limit.

What remains possible

The obstruction is specific to the current table and a uniform forward horizon. It leaves open a proof that each candidate eventually reaches a useful rule after a seed-dependent number of steps. New rules may also break the obstruction. The constructed seeds have long noncontracting prefixes, but they are not proved to be fully noncontracting or divergent. This is a limitation of one proof approach, not a counterexample to Collatz.

Where to inspect the evidence

The formal module is `lean/Collatz/MergeRuleObstruction.lean`. From `lean/`, run `lake build Collatz.MergeRuleObstruction`. From the root, run `python3 analysis/merge_rule_obstruction.py --horizon 64`. The adjacent test file checks residue agreement with Lean, actual orbit steps, and the available rule outputs.